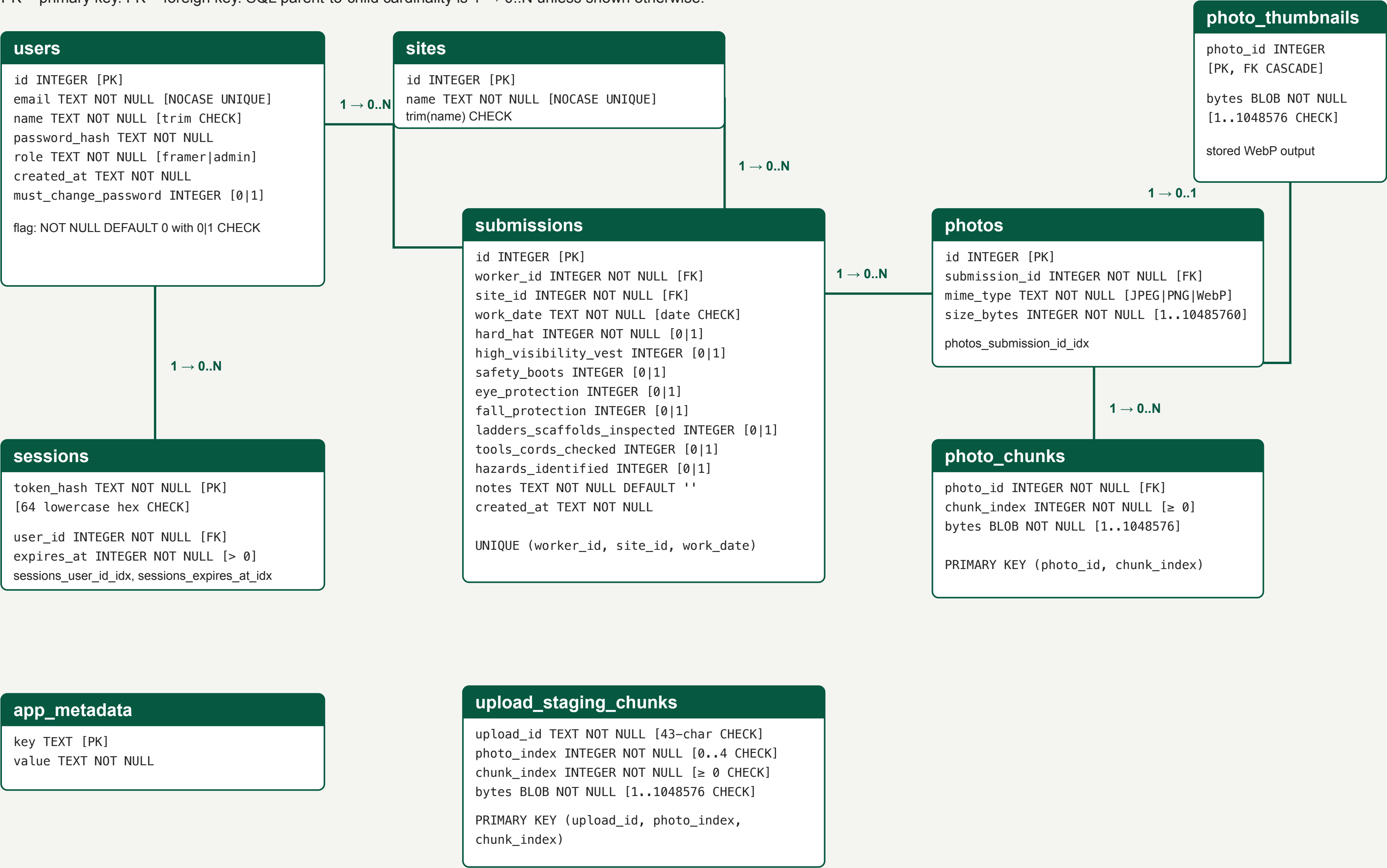


RAS Site Safety — SQLite Durable Object ERD

Nine STRICT tables in SAFETY_APP / ras-site-safety. This diagram is simplified. See server/schema.sql for every NOT NULL and CHECK detail.

PK = primary key. FK = foreign key. SQL parent-to-child cardinality is 1 → 0..N unless shown otherwise.



API invariants: one to five decoded photos per submission. Original bytes use contiguous chunks of at most one MiB.

The unique worker/site/date key prevents duplicate submissions. The bootstrap marker prevents reseeding.

Cascade direction: deleting a photos row deletes matching photo_chunks and photo_thumbnails rows.

upload_staging_chunks is temporary, private, request-scoped, and cleared after failure or commit.

Indexes: photos_submission_id_idx, sessions_user_id_idx, sessions_expires_at_idx.

Application checks add Gregorian dates, ownership, photo count, dimensions, pixels, quotas, and session expiry.

Thumbnail rows have a one-to-zero-or-one relationship with photos and store bounded WebP bytes.

Migration: add users.must_change_password when absent, then record schema_version=3 in app_metadata.

FKs: submissions.worker_id → users.id. submissions.site_id → sites.id. photos.submission_id → submissions.id.

FKs: sessions.user_id → users.id. photo_chunks.photo_id and photo_thumbnails.photo_id → photos.id.